

# Anant A. Joshi

✉ anantaj2@illinois.edu | 🌐 <https://anantjoshi97.github.io/website> | 📧 anantjoshi97 | 📄 anantjoshi97 | 🎓 Google Scholar

## Summary

- As a control theorist and applied mathematician, my foundation is in **stochastic differential equations, stochastic optimal control (SOC), and filtering**. I am interested in problems in **sampling, SOC, generative models, and optimization**. My research focusses on **constructing sampling-based algorithms for SOC and reinforcement learning (RL) using interacting particle systems**.

## Education

**Ph.D. in Mech. Engg., Univ. of Illinois Urbana-Champaign (GPA 4.0/4.0, Advisor: Prof. Prashant Mehta)**

Jan. 2021 - present

- Awarded: Grainger Engineering Graduate Fellowship

**B.Tech. in Mech. Engg. and M.Tech. in Systems and Control Engg., IIT Bombay, India (GPA 9.44/10)**

Aug. 2015 - July 2020

- Awarded: Best student paper award, 29th AAS/AIAA Space-Flight Mechanics Meeting, 2019

## Patents

- S. Mowlavi, M. Benosman, **A. A. Joshi**, "System and Method Suitable for Synthesizing a Control Policy for a Mechanical System", USPTO App. 19/273,383

## Experience

### dual-EnKF Algorithm For Fast Simulation-Based Reinforcement Learning

UIUC

RESEARCH WITH PROF. P. G. MEHTA, PROF. A. TAGHVAEI, PROF. S. P. MEYN

Jan 2021 - Dec 2024

- Developed a simulation based algorithm (dual-EnKF) for stochastic optimal control and RL in continuous-time continuous-state systems.
- Represented value function as a probability density and deployed the Ensemble Kalman Filter (EnKF) to sample efficiently from it.
- Numerically benchmarked dual-EnKF against policy gradient method, demonstrating upto 100x computational speedup.

### Industry Internship: dual-EnKF for Data Driven Robust Control of PDEs

Mitsubishi Electric Research Labs

RESEARCH INTERN WITH DR. MOUHACINE BENOSMAN AND DR. SAVIZ MOWLAVI

June 2024 - August 2024

- Developed a data-driven, robust control algorithm for nonlinear systems governed by partial differential equations. Combined an optimal control estimated using dual-EnKF with a robust control obtained from Lyapunov redesign.
- Validated the controller's performance and robustness through extensive simulations on the heat equation and Burgers' equation.

### Error Analysis of Sampling Algorithms for Stochastic Control

UIUC

RESEARCH WITH PROF. P. G. MEHTA AND PROF. A. TAGHVAEI

Jan 2025 - ongoing

- Developed an interacting particle-based sampling algorithm to numerically solve discrete-time stochastic optimal control. Analyzed the algorithm's approximation error and performance, and benchmarked it against traditional importance sampling methods.
- Highlighted scalability by demonstrating that the proposed algorithm avoids "curse of dimensionality" seen in importance sampling.

### Observer Design for Systems on Smooth Manifolds with Lie Group Symmetry

IIT Bombay, India

RESEARCH PROJECT WITH PROF. R. N. BANAVAR AND PROF. D. H. S. MAITHRIPALA

Aug. 2018 - Jul. 2019

- Leveraged inherent geometric symmetry in nonlinear mechanical systems to simplify design observer into two smaller subsystems.

### Uncertainty Quantification in Dynamical Systems with Invariants (AAS/AIAA Best paper)

University of Texas at Arlington

RESEARCH INTERNSHIP WITH PROF. K. S. SUBBARAO

May 2018 - July 2018

- Modeled impact of Gaussian white noise on dynamical systems to quantify uncertainty in conserved quantities (invariants).
- Characterized the temporal evolution of the statistical moments of invariants. Theoretical results were verified with Monte Carlo simulation.

## Leadership Roles

### Head, Attitude Determination and Control Subsystem, Student Satellite Project

IIT Bombay

- Responsible for envisioning and executing technical work in the 10 member subsystem. Regularly presented progress review to the faculty mentors.
- Laid out Quality-Assurance practices to ensure categorical documentation of project progress.

## Selected Publications

- A. A. Joshi**, S. Mowlavi, M. Benosman, "A Dual Ensemble Kalman Filter Approach to Robust Control of Nonlinear Systems: An Application to Partial Differential Equations", accepted to IEEE Conference on Decision and Control (CDC), 2025
- A. A. Joshi**, H.-S. Chang, A. Taghvaei, P. G. Mehta, S. P. Meyn, "Interacting Particle Systems for Fast Linear Quadratic RL", Learning for Dynamics & Control Conference (L4DC), 2024
- P. Katdare, **A. A. Joshi**, K. Driggs-Campbell, "Towards Provable Log Density Policy Gradient", Transactions on Machine Learning Research (TMLR), 2024
- A. A. Joshi**, D. Chatterjee, R. N. Banavar, "Robust Discrete-Time Pontryagin Maximum Principle on Matrix Lie Groups", IEEE Transactions on Automatic Control, 2021
- A. A. Joshi**, K. Subbarao, "Uncertainty Quantification and Analysis of Dynamical Systems with Invariants", AAS-19-368, 29th AAS/AIAA Space-Flight Mechanics Meeting, January, 2019

## Technical Skills

**PROGRAMMING:** Python, PyTorch, MATLAB (Simulink, Control System Toolbox, Optimization Toolbox), C++

**CONTROL & ESTIMATION:** PID, LQR, Model Predictive Control (MPC), Sensor Fusion, State Estimation, Kalman Filters (EnKF, EKF, UKF)

**COURSEWORK:** Optimization, Reinforcement Learning, Machine Learning for Signal Processing, Deep Generative Models, State Estimation, Adaptive Control, Optimal Control, Stochastic Control, Probability and Random Processes, Stochastic Differential Equations